

ZODENOX[®] FILM COATED TABLETS 10MG

Ingredient(s):

Each film-coated tablet contains:

Zolpidem Tartrate 10mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

Zolpidem tartrate is an imidazopyridine which preferentially binds the omega-1 receptor subtype (also known as the benzodiazepine-1 subtype) which corresponds to GABA-A receptors containing the alpha-1 sub-unit, whereas benzodiazepines non-selectively bind both omega-1 and omega-2 subtypes. The modulation of the chloride anion channel via this receptor leads to the specific sedative effects demonstrated by zolpidem tartrate. These effects are reversed by the benzodiazepine antagonist flumazenil.

Zolpidem tartrate decreases sleep latency and the number of awakenings, and increases sleep duration and sleep quality. These effects are associated with a characteristic EEG profile, different from that of the benzodiazepines. In studies that measured the percentage of time spent in each sleep stage, zolpidem tartrate has generally been shown to preserve sleep stages. At the recommended dose, zolpidem tartrate has no influence on the paradoxical sleep duration (REM). The preservation of deep sleep (stages 3 and 4 - slow-wave sleep) may be explained by the selective omega-1 binding by zolpidem tartrate. All identified effects of zolpidem tartrate are reversed by the benzodiazepine antagonist flumazenil.

Pharmacokinetic:

Zolpidem tartrate has both a rapid absorption and onset of hypnotic action. Bioavailability is 70% following oral administration and demonstrates linear kinetics in the therapeutic dose range. Peak plasma concentration is reached at between 0.5 and 3 hours.

The elimination half-life is short, with a mean of 2.4 hours (± 0.2 h) and a duration of action of up to 6 hours.

Protein binding amounts to $92.5\% \pm 0.1\%$. First pass metabolism by the liver amounts to approximately 35%. Repeated administration has been shown not to modify protein binding indicating a lack of competition between zolpidem tartrate and its metabolites for binding sites.

The distribution volume in adults is 0.54 ± 0.02 L/kg and decreases to 0.34 ± 0.05 L/kg in the very elderly.

All metabolites are pharmacologically inactive and are eliminated in the urine (56%) and in the faeces (37%).

Zolpidem tartrate has been shown in trials to be non-dialysable.

Plasma concentrations in elderly subjects and those with hepatic impairment are increased. In patients with renal insufficiency, whether dialysed or not, there is a moderate reduction in clearance. The other pharmacokinetic parameters are unaffected.

Zolpidem tartrate is metabolised via several hepatic cytochrome P450 enzymes, the main enzyme being CYP3A4 with the contribution of CYP1A2. Since CYP3A4 plays an important role in zolpidem tartrate metabolism, possible interactions with drugs that are substrates or inducers of CYP3A4 should be considered.

Indication(s):

Occasional and transient insomnia.

Dosage and Administration:

The treatment should always be implemented at the lowest effective dose and maximum dosage never exceeded.

1. The usual dose for adults is one 10-mg tablet daily. Zolpidem should always be taken just before going to bed.
2. In elderly patients or patients with hepatic insufficiency: Dosage should be halved, i.e. 5 mg. Dosage must never exceed 10 mg/day.

Safety and effectiveness of zolpidem in paediatric patients <18 years have not been established. Therefore, zolpidem should not be prescribed in this population.

Oral administration

Contraindications:

1. Hypersensitivity to Zolpidem.
2. Severe respiratory insufficiency, sleep apnoea syndrome, severe, acute or chronic hepatic insufficiency (risk of encephalopathy), myasthenia.
3. Due to the lactose content of this product, it is contraindicated in the event of congenital galactosaemia, glucose or galactose malabsorption syndrome or lactase deficiency.

Precaution(s) / Warning(s):

1. Respiratory Insufficiency: As hypnotics have the capacity to depress respiratory drive, precautions should be observed if zolpidem is prescribed to patients with compromised respiratory function.
2. Hepatic Insufficiency: As clearance and metabolism of zolpidem tartrate is reduced in hepatic impairment, dosage should begin at 5mg in these patients with particular caution being exercised in elderly patients. In adults (under 65 years) dosage may be increased to 10mg only where the clinical response is inadequate and the drug is well tolerated.
3. The cause of insomnia should be identified wherever possible and the underlying factors treated before a hypnotic is prescribed. The failure of insomnia to remit after a 7-14 day course of treatment may indicate the presence of a primary psychiatric or physical disorder, and the patient should be carefully re-evaluated at regular intervals.
4. Elderly: Elderly or debilitated patients may be especially sensitive to the effects of zolpidem tartrate therefore a 5mg dose is recommended. These recommended doses should not be exceeded.
5. Paediatric Patients: Safety and effectiveness of zolpidem have not been established in patients below the age of 18 years
6. Depression: zolpidem tartrate should be administered with caution in patients exhibiting symptoms of depression. Suicidal tendencies may be present therefore the least amount of zolpidem that is feasible should be supplied to these patients to avoid the possibility of intentional overdosage by the patient. Pre-existing depression may be unmasked during use of zolpidem. Since insomnia may be a symptom of depression, the patient should be re-evaluated if insomnia persists.
7. Extreme caution should be exercised when prescribing for patients with a history of drug or alcohol abuse since they are at risk of habituation and psychological dependence.
8. General information relating to effects seen following administration of benzodiazepines and other hypnotic agents which should be taken into account by the prescribing physician are described below.
 - a) Tolerance: Some loss of efficacy to the hypnotic effects of short-acting benzodiazepines and benzodiazepine-like agents like zolpidem may develop after repeated use for a few weeks.
 - b) Dependence: Use of benzodiazepines or benzodiazepine-like agents like zolpidem may lead to the development of physical and psychological dependence. The risk of dependence increases with dose and duration of treatment; it is also greater in patients with a history of psychiatric disorders and/or alcohol or drug abuse. These patients should be under careful surveillance when receiving hypnotics. Once physical dependence has developed, abrupt termination of treatment will be accompanied by withdrawal symptoms. These may consist of headaches or muscle pain, extreme anxiety and tension, restlessness, confusion and irritability. In severe cases the following symptoms may occur: derealisation, depersonalisation, hyperacusis, numbness and tingling of the extremities, hypersensitivity to light, noise and physical contact, hallucinations or epileptic seizures.
 - c) Rebound insomnia :A transient syndrome whereby the symptoms that led to treatment with a benzodiazepine or benzodiazepine-like agent recur in an enhanced form, may occur on withdrawal of hypnotic treatment. It may be accompanied by other reactions including mood changes, anxiety and restlessness.
 - d) Amnesia: Benzodiazepines or benzodiazepine-like agents such as zolpidem may induce anterograde amnesia. The condition occurs most often several hours after ingesting the product and therefore to reduce the risk patients should ensure that they will be able to have an uninterrupted sleep of 7-8 hours.
 - e) Other psychiatric and "paradoxical" reactions: Other psychiatric and paradoxical reactions like restlessness, exacerbated insomnia, agitation, irritability, aggression, delusion, anger, nightmares, hallucinations, psychosis, abnormal behaviour and other adverse behavioural effects are known to occur when using benzodiazepines or benzodiazepine-like agents. Should this occur, use of the product should be discontinued. These reactions are more likely to occur in the elderly.

- f) Somnambulism and associated behaviours: Sleep walking and other associated behaviours such as "sleep driving", preparing and eating food, making phone calls or having sex, with amnesia for the event, have been reported in patients who had taken zolpidem and were not fully awake. The use of alcohol and other CNS-depressants with zolpidem appears to increase the risk of such behaviours, as does the use of zolpidem at doses exceeding the maximum recommended dose. Discontinuation of zolpidem should be strongly considered for patients who report such behaviours (for example, sleep driving), due to the risk to patients and others.
9. Effects on ability to drive or use machines: Drivers and machine operators should be informed of the risk of drowsiness associated with the use of this product.

WARNING

1. Anaphylaxis (severe allergic reaction) and angioedema (severe facial swelling) which can occur as early as the first time the product is taken.
2. Complex sleep related behaviours can occur which may include sleep driving, making phone calls, preparing and eating food while sleeping.

Interaction with Other Medicaments

1. Not recommended : Concomitant intake with alcohol. The sedative effect may be enhanced when the product is used in combination with alcohol. This affects the ability to drive or use machines.
2. Take into account: Combination with CNS depressants. Enhancement of the central depressive effect may occur in cases of concomitant use with antipsychotics (neuroleptics), hypnotics, anxiolytics/sedatives, antidepressant agents, narcotic analgesics, antiepileptic drugs, anaesthetics and sedative antihistamines. Zolpidem tartrate appears to interact with sertraline. This interaction may cause increased drowsiness. Also, isolated cases of visual hallucinations were reported. In the case of narcotic analgesics enhancement of euphoria may also occur leading to an increase in psychological dependence.
3. CYP450 Inhibitors: Compounds which inhibit/induce certain hepatic enzymes (particularly cytochrome P450) may enhance/reduce the activity of benzodiazepines and benzodiazepine-like agents.
4. Other drugs: When zolpidem tartrate was administered with ranitidine or cimetidine, no significant pharmacokinetic interactions were observed.

Pregnancy and Lactation

Although animal studies have shown no teratogenic or embryotoxic effects, safety in pregnancy has not been established. As with all drugs zolpidem tartrate should be avoided in pregnancy particularly during the first trimester.

If the product is prescribed to a woman of childbearing potential, she should be warned to contact her physician about stopping the product if she intends to become or suspects that she is pregnant.

If, for compelling medical reasons, zolpidem tartrate is administered during the late phase of pregnancy, or during labour, effects on the neonate, such as hypothermia, hypotonia and moderate respiratory depression, can be expected due to the pharmacological action of the product. Cases of severe neonatal respiratory depression have been reported when zolpidem tartrate was used with other CNS depressants at the end of pregnancy.

Infants born to mothers who took benzodiazepines or benzodiazepine-like agents chronically during the latter stages of pregnancy may have developed physical dependence and may be at some risk of developing withdrawal symptoms in the postnatal period.

Small quantities of zolpidem tartrate appear in breast milk. The use of zolpidem tartrate in nursing mothers is therefore not recommended.

Side Effect(s) / Adverse Reaction(s):

These depend on dosage and the patient's individual sensitivity.

1. Undesirable Neuro-Psychiatric Effects: Anterograde amnesia may occur at therapeutic dose levels with the risk increasing in proportion to dose; behaviour disorders, impaired consciousness, irritability, aggressiveness, agitation, hallucinations, somnambulism; physical and psychic drug dependence, even at therapeutic dose levels accompanied by withdrawal symptoms or rebound insomnia when treatment is discontinued; feelings of ebriety, headache, ataxia; confusion, reduced vigilance even drowsiness (especially in elderly patients), insomnia, nightmares, tension; changes in libido.
2. Undesirable Cutaneous Effects: Skin rash , pruritus (itching), superficial or deep (angioedema) urticaria, hyperhidrosis.
3. Undesirable General Effects: Muscle hypotonia, asthenia, unsteady gait or falls (especially in elderly subjects when dosage recommendations for zolpidem are not followed).
4. Undesirable Ocular Effects: Diplopia.
5. Undesirable Gastrointestinal Effects: Gastrointestinal tract disorders: Diarrhoea, nausea, vomiting, abdominal pain.
6. Hepatic Effects: Elevated liver enzymes.
7. Immune System Effects: Angioedema (eg, Quincke's edema).

Symptoms and Treatment for Overdosage, and Antidote(s):

Signs and Symptoms:

In cases of overdose involving zolpidem tartrate alone or with other CNS-depressant agents (including alcohol), impairment of consciousness ranging from somnolence to coma and including fatal outcomes have been reported.

Management:

General symptomatic and supportive measures should be used. If there is no advantage in emptying the stomach, activated charcoal should be given to reduce absorption. Sedating drugs should be withheld even if excitation occurs.

Use of flumazenil may be considered where serious symptoms are observed.

Flumazenil is reported to have an elimination half-life of about 40 to 80 minutes. Patients should be kept under close observation because of this short duration of action; further doses of flumazenil may be necessary. However, flumazenil administration may contribute to the appearance of neurological symptoms (convulsions). The value of dialysis in the treatment of an overdose has not been determined. Dialysis in patients with renal failure receiving therapeutic doses of zolpidem have demonstrated no reduction in levels of zolpidem.

In the management of overdose with any medicinal product, it should be borne in mind that multiple agents may have been taken

Shelf-Life:

2 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C. Protect from light and moisture.

Product Description and Packing(s):

A white round film coated tablet, one side impressed with "YSP" on the top, "173" on the bottom. Blister packing of 10's x 3, 10's x 10, 10's x 50 and 10's x 100.



Y. S. P.

Acknowledgement

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